Anwar Mousa

Data Engineer

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Summary

Data Engineer with hands-on experience building ETL/ELT pipelines, real-time data systems, analytics platforms, and cloud-based data solutions. Experienced with **Python, SQL, Apache Airflow, Kafka, Spark, Snowflake, PostgreSQL, and dbt**, with a strong foundation in data modeling, data quality, orchestration, and modern data architecture. Also experienced in building AI-powered systems where data pipelines, observability, and analytics support intelligent applications.

Experience

Data & AI Systems Engineer *Present*
ThinkerLab · Egypt

- Build data-driven software systems and analytics solutions for real-world business use cases, combining data engineering, backend services, and AI capabilities.
- Design and implement data workflows, APIs, and production-oriented systems with a focus on reliability, maintainability, and practical business requirements.
- Integrate modern data and AI tooling across analytics platforms, automation workflows, and SaaS products.

BI Engineer *Jul 2025 – Jan 2026*
Link Development · Hybrid

- Developed and maintained business intelligence solutions, dashboards, and analytical reporting workflows for organizational stakeholders.
- Built and optimized data models and ETL workflows to improve reporting reliability and reduce data processing latency.
- Translated business requirements into analytical solutions while maintaining data quality and consistency across reporting outputs.

Data Analytics *Oct 2024 – May 2025*
Digital Egypt Pioneers Initiative (DEPI) – MCIT · Egypt

- Completed an intensive, project-based national program covering data analytics, ETL pipeline design, Apache data technologies, and business intelligence.
- Designed and implemented ETL pipelines using Apache Airflow, reducing data processing time by 40
- Optimized data warehouse queries in Snowflake, improving query performance by 35redesign and clustering keys.
- Collaborated with cross-functional teams to deliver data-driven insights supporting strategic business decisions.

Education

Bachelor of Computer Science and Informatics *Aug 2022 – Jun 2026*
Delta University for Science and Technology · Egypt

- Graduation Project: **Ayn** — AI-native QA & accreditation platform for educational institutions Demo

Selected Projects

DataOps Agent – Data Observability & Remediation GitHub — Demo

- Built an agentic data operations platform combining **dlt**, **PostgreSQL**, **dbt**, and **Dagster** for ingestion, transformation, orchestration, and automated data quality monitoring.
- Implemented **27 dbt test assertions**, Dagster asset checks, and a **22-tool FastMCP** interface for structured investigation and remediation workflows.
- Designed a human-in-the-loop governance flow with approval gates, auditability, and controlled remediation instead of unrestricted automated changes.

ResolveAI – AI Support Ticket Triage & Auto-Resolution GitHub

- Built the data layer for an AI-powered support automation system using **PostgreSQL**, **dbt**, and a structured analytical warehouse model.
- Implemented **Apache Airflow** orchestration and data workflows supporting ticket ingestion, analytics, evaluation, and operational metrics.
- Integrated **pgvector** and evaluation workflows to support grounded retrieval and measurable AI system performance.

Real-Time Forex Data Pipeline & Analytics Dashboard GitHub

- Built a real-time ETL pipeline using **Apache Airflow, Kafka, and Spark** to process currency exchange data from the Alpha Vantage API.
- Designed a **Snowflake** warehouse using Bronze/Silver/Gold medallion architecture with automated data quality checks.
- Containerized the pipeline with **Docker** and built a Streamlit analytics layer for technical indicators, volatility analysis, and data export.

NASA Log Analytics Pipeline GitHub

- Processed **3.4M+ HTTP log entries and 61+ GB** using an end-to-end pipeline built with **Apache Airflow, Spark, and PostgreSQL**.
- Applied anomaly detection to identify traffic deviations and error-rate spikes and visualized operational metrics through **Grafana**.

Certifications

- **AWS Certified Cloud Practitioner** — Amazon Web Services
- **Google Data Analytics Professional Certificate** — Google
- **Data Analytics – Google Data Analyst Specialist** — Digital Egypt Pioneers Initiative

Technical Skills

Languages:	Python, SQL, PySpark, TypeScript
Data Engineering:	Apache Airflow, Apache Kafka, Apache Spark, dbt, ETL/ELT
Data Platforms:	Snowflake, PostgreSQL, Delta Lake, AWS S3, Data Warehousing
Data Architecture:	Data Modeling, Dimensional Modeling, Medallion Architecture, Data Quality
Analytics:	Power BI, Streamlit, Grafana
AI & Backend:	FastAPI, LLM Orchestration, RAG, API Integration
Engineering:	Docker, Git, GitHub, Linux, Bash